

Assignment

1. Classification model:

a. Loan Status Prediction:

- i. This is a Property Loan Status Prediction dataset, we have the data of applicants who had previously applied for the loan based on the property and applicant-based metrics.
- ii. The bank will decide whether to give a loan to the applicant based on some factors such as Applicant Income, Loan Amount, previous Credit History, Co-applicant Income, etc.
- iii. The objective is to build a Machine Learning Model to predict the loan to be approved or to be rejected for an applicant.
- iv. You are free to use any classification model such Logistic Regression, Decision Tree, RF or XGBoost.
- v. Write the performance and key insights on the data in a word doc.

2. Regression model:

a. Sales Forecasting:

- i. You are provided with historical sales data for 45 retail stores located in different regions. Each store contains a number of departments, and you are tasked with predicting the department-wide sales for each store.
- ii. In addition, the retail company runs several promotional markdown events throughout the year. These markdowns precede prominent holidays, the four largest of which are the Super Bowl, Labour Day, Thanksgiving, and Christmas.
- iii. Data:
 1. stores.csv: This file contains anonymized information about the 45 stores, indicating the type and size of the store.
 2. features.csv: Comprise of information on Markdown and macroeconomic factors.
 3. train.csv: This is the historical training data.
- iv. Write the performance and key insights on the data in a word doc.

3. Clustering:

a. Credit Card Data: Marketing Strategy

- i. This case requires to develop a customer segmentation to define marketing strategy.
- ii. Sample Dataset summarizes the usage behaviour of about 9000 active credit card holders during the last 6 months. The file is at a customer level with 18 behavioural variables.

iii. Following is the Data Dictionary for Credit Card dataset:

CUST_ID: Identification of Credit Card holder (Categorical)

BALANCE: Balance amount left in their account to make purchases

BALANCE_FREQUENCY: How frequently the Balance is updated, score between 0 and 1
(1 = frequently updated, 0 = not frequently updated)

PURCHASES: Amount of purchases made from account

ONEOFF_PURCHASES: Maximum purchase amount done in one-go

INSTALLMENTS_PURCHASES: Amount of purchase done in installment

CASH_ADVANCE: Cash in advance given by the user

PURCHASES_FREQUENCY: How frequently the Purchases are being made, score between 0 and 1 (1 = frequently purchased, 0 = not frequently purchased)

ONEOFFPURCHASESFREQUENCY: How frequently Purchases are happening in one-go
(1 = frequently purchased, 0 = not frequently purchased)

PURCHASESINSTALLMENTSFREQUENCY: How frequently purchases in installments are being done (1 = frequently done, 0 = not frequently done)

CASHADVANCEFREQUENCY: How frequently the cash in advance being paid

CASHADVANCETRX: Number of Transactions made with "Cash in Advanced"

PURCHASES_TRX: Number of purchase transactions made

CREDIT_LIMIT: Limit of Credit Card for user

PAYMENTS: Amount of Payment done by user

MINIMUM_PAYMENTS: Minimum amount of payments made by user

PRCFULLPAYMENT: Percent of full payment paid by user

TENURE: Tenure of credit card service for user